

Rewriting Numbers in Equivalent FormsAdditional Practice 7A.01.01 **ANSWER KEY**

1. Which strategy will always work to rewrite a fraction as a decimal? Fill in one bubble.

- ☐ Multiply or divide by 100.
- ☒ **Divide the numerator by the denominator.**
- ☐ Create an equivalent fraction with a denominator of 100.
- ☐ Use the place value of the last digit to determine the denominator.

Dividing always works. Making an equivalent fraction with a denominator of 100 works for $17/20$, but not for $1/3$ — no whole number times 3 equals 100.

2. Which strategy will always work to rewrite a terminating decimal as a fraction? Fill in one bubble.

- ☐ Multiply or divide by 100.
- ☐ Divide the numerator by the denominator.
- ☐ Create an equivalent fraction with a denominator of 100.
- ☒ **Use the place value of the last digit to determine the denominator.**

In 0.85 the last digit is in the hundredths place, so the denominator is 100 and the fraction is $85/100$, which simplifies to $17/20$.

For questions 3 through 5, write each fraction as an equivalent decimal. Use bar notation for any decimal that repeats. Then fill in one bubble.

3. $\frac{27}{8} = 3.375$

This is a ☒ **terminating decimal** ☐ repeating decimal.

4. $\frac{6}{11} = 0.\overline{54}$

This is a ☐ terminating decimal ☒ **repeating decimal.**

5. $\frac{4}{5} = 0.8$

This is a ☒ **terminating decimal** ☐ repeating decimal.

$27 \div 8 = 3.375$ exactly. $6 \div 11 = 0.545454\dots$, so both digits repeat. $4 \div 5 = 0.8$ exactly.

6. The decimal equivalent of $59/11$ is $5.3636\dots$. Which of these is $5.3636\dots$ written in bar notation? Fill in one bubble.

- ☐ A. $5.\overline{36}$
- ☒ B. $5.\overline{36}$
- ☐ C. $5.3\overline{6}$
- ☐ D. 5.36

Both the 3 and the 6 repeat: $59 \div 11 = 5.363636\dots$, so the bar covers 36. C would mean 5.36666... D is 5.36 exactly. A puts a bar over the whole number, which bar notation never does.

7. $\frac{5}{11} = 0.\overline{45}$

$5 \div 11 = 0.454545\dots$

Use your answer to question 7 for questions 8 through 10. Predict each decimal equivalent, then check it with your calculator. Use bar notation for every repeating decimal.

8. $2\frac{5}{11} = 2.\overline{45}$

9. $\frac{10}{11} = 0.\overline{90}$

In elevenths the two repeating digits come from the numerator. $5/11 = 0.4545\dots$, so $2\frac{5}{11} = 2.4545\dots$ $10/11 = 0.909090\dots$, so the bar covers 90.

10. Use $87/11$ for parts a and b.

a. Write $87/11$ as a mixed number. $7\frac{10}{11}$

b. Write $87/11$ as a decimal. Use bar notation. $7.\overline{90}$

$87 \div 11 = 7$ with a remainder of 10, so $87/11 = 7\frac{10}{11}$. From question 9, $10/11 = 0.909090\dots$, so the answer is $7.909090\dots$

11. Dominic divides 1 by 6 on his calculator. The screen shows 0.166666667 .

a. Is $1/6$ a terminating decimal or a repeating decimal? Fill in one bubble.

☐ Terminating.

☒ **Repeating.**

b. Write $1/6$ in bar notation. $0.1\overline{6}$

c. Explain why the last digit on the screen is a 7.

The calculator has room for only ten digits, so it rounds the last one it shows. The true value is $0.1666\dots$ and it never stops, so the 7 on the screen is the calculator rounding, not part of the pattern.

Read the pattern from the middle of the display, never from the last digit.